

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – MCQ Test

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO3 – Precision (BL3, BL5)	Assessment:	Assessment Tool 3 – MCQ Test
Weightage:	30% of CIA	Total Marks:	100
CO3 Statement:	Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, BTrees, Red-Black Trees, and Splay Trees.		
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Section / Batch:	AIDS	Date:	10/08/2026

Code of Conduct

I Jeeva Raghul .M (192524487) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: Jeeva Raghul .M

Instructions

- This test contains 25 MCQ Test questions (4 marks each), Total: 100 Marks.
- When a student answer using MCQ, in evaluation the answer should be with following requirements:
Identify the most appropriate answer, conceptual understanding, problem-solving ability, application of knowledge.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Preliminaries – Tree Terminology	Root, height, level, siblings and sub tree	1–2	8
Binary Tree, Binary Search Tree	Properties, binary tree and binary search tree, insertion and deletion	3–9	28
Tree Traversals	Traversal types, In order, Post order and Pre order	10–14	20
AVL Tree	Balancing factor, Rotation, Types of rotation, examples	15–16	8
Applications of Search Trees	Usage of binary tree	17	4
B Tree - 2-3 tree, 2-3-4 tree	Properties, Example and Operations	18–19	8
TRIE	Properties, structures and Operations	20	4
Red Black Tree	Properties, Balancing and Operations	21–22	8

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QUESTIONS

Q1 ✓
Tree Terminology
4 marks**Q2** ✓
Quad Tree
4 marks**Q3** ✓
Splay Tree
4 marks**Q4** ✓
Splay Trees
4 marks**Q5** ✓
Splay Tree
4 marks**Q6** ✓
Red-Black Tree
4 marks**Q7** ✓
Red-Black Tree**Q1 Tree Terminology**

4 marks

Nodes having the same parent are called:

☐ A. Ancestors☐ B. Descendants☒ C. Siblings☐ D. Leaves Save Answer

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QUESTIONS

Q1
Tree Terminology
4 marks**Q2**
Quad Tree
4 marks**Q3**
Splay Tree
4 marks**Q4**
Splay Trees
4 marks**Q5**
Splay Tree
4 marks**Q6**
Red-Black Tree
4 marks**Q7****Q2 Quad Tree**

4 marks

In a Quad Tree, each internal node has at most:

A. 2 children**B.** 3 children**C.** 4 children**D.** 8 children Save Answer

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QUESTIONS

Q1 ✓
Tree Terminology
4 marks**Q2** ✓
Quad Tree
4 marks**Q3** ✓
Splay Tree
4 marks**Q4** ✓
Splay Trees
4 marks**Q5** ✓
Splay Tree
4 marks**Q6** ✓**Q3 Splay Tree**

4 marks

After accessing a node in a Splay Tree, that node is moved to:

A. Leaf

B. Middle

C. Root

D. NULL

 Save Answer

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QUESTIONS

Q1 ✓
Tree Terminology
4 marks**Q2** ✓
Quad Tree
4 marks**Q3** ✓
Splay Tree
4 marks**Q4** ✓
Splay Trees
4 marks**Q5** ✓
Splay Tree
4 marks**Q6** ✓
Red-Black Tree
4 marks**Q7** ✓
Red-Black Tree
4 marks**Q4 Splay Trees**

4 marks

Identify the rotation:

```
50
 \
 70
 \
 90
```

☐ A. Zag-Zag☒ B. Zig-Zig☐ C. Zig☐ D. AVL Rotation Save Answer

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QUESTIONS

Q1 ✓
Tree Terminology
4 marks**Q2** ✓
Quad Tree
4 marks**Q3** ✓
Splay Tree
4 marks**Q4** ✓
Splay Trees
4 marks**Q5** ✓
Splay Tree
4 marks**Q6** ✓
Red-Black Tree
4 marks**Q5 Splay Tree**

4 marks

Splay Tree is a type of:

- A. Complete Binary Tree
- B. Self-adjusting Binary Search Tree**
- C. Heap Tree
- D. Graph

 Save Answer

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QUESTIONS

Q1 ✓
Tree Terminology
4 marks**Q2** ✓
Quad Tree
4 marks**Q3** ✓
Splay Tree
4 marks**Q4** ✓
Splay Trees
4 marks**Q5** ✓
Splay Tree
4 marks**Q6** ✓**Q6 Red-Black Tree**

4 marks

If a red node has a red child after insertion, what is this called?

A. Black-height problem**B.** BST violation**C.** Root violation**D.** Red-Red violation Save Answer

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QUESTIONS

Q1 ✓
Tree Terminology
4 marks**Q2** ✓
Quad Tree
4 marks**Q3** ✓
Splay Tree
4 marks**Q4** ✓
Splay Trees
4 marks**Q5** ✓
Splay Tree
4 marks**Q6** ✓
Red-Black Tree
4 marks**Q7 Red-Black Tree**

4 marks

Insert sequence 70, 50, 90, 40 causes insertion of 40 under red parent. Uncle is red. What is needed?

A. Rotation

B. Recoloring

C. Deletion

D. No change

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QUESTIONS

Q1 ✓
Tree Terminology
4 marks**Q2** ✓
Quad Tree
4 marks**Q3** ✓
Splay Tree
4 marks**Q4** ✓
Splay Trees
4 marks**Q5** ✓
Splay Tree
4 marks**Q6** ✓**Q8 Red-Black Tree**

4 marks

If black height differs in two paths from the same node, the tree becomes:

☐ A. Valid☒ B. Invalid☐ C. Balanced☐ D. Complete Save Answer

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QUESTIONS

Q1 ✓
Tree Terminology
4 marks**Q2** ✓
Quad Tree
4 marks**Q3** ✓
Splay Tree
4 marks**Q4** ✓
Splay Trees
4 marks**Q5** ✓
Splay Tree
4 marks**Q6** ✓
Red-Black Tree
4 marks**Q9 Red-Black Tree**

4 marks

Every path from a node to its descendant NULL nodes must have the same number of

A. Black nodes**B.** Red nodes**C.** Leaf nodes**D.** Parent nodes Save Answer

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QUESTIONS

Q1
Tree Terminology
4 marks**Q2**
Quad Tree
4 marks**Q3**
Splay Tree
4 marks**Q4**
Splay Trees
4 marks**Q5**
Splay Tree
4 marks**Q6**
Red-Black Tree**Q10 2-3 Tree**

4 marks

In a balanced 2-3 Tree, all leaves must be at:

A. Different heights**B.** Root level**C.** Random height**D.** Same height Save Answer

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QUESTIONS

Q1 ✓
Tree Terminology
4 marks**Q2** ✓
Quad Tree
4 marks**Q3** ✓
Splay Tree
4 marks**Q4** ✓
Splay Trees
4 marks**Q5** ✓
Splay Tree
4 marks**Q6** ✓
Red-Black Tree
4 marks**Q7** ✓
Red-Black Tree
4 marks**Q11 2-3-4 Tree**

4 marks

Insert 10, 20, 30, 40 in a 2-3-4 Tree. What happens after inserting 40?

A. Tree becomes unbalanced**B.** Node split may occur**C.** Node is deleted**D.** No change Save Answer

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QUESTIONS

Q1 ✓
Tree Terminology
4 marks**Q2** ✓
Quad Tree
4 marks**Q3** ✓
Splay Tree
4 marks**Q4** ✓
Splay Trees
4 marks**Q5** ✓
Splay Tree
4 marks**Q6** ✓
Red-Black Tree
4 marks**Q7** ✓
Red-Black Tree**Q12 2-3 Tree**

4 marks

The maximum number of children in a 2-3 Tree node is:

A. 3**B.** 2**C.** 5**D.** 4 Save Answer

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QUESTIONS

Q1 ✓
Tree Terminology
4 marks**Q2** ✓
Quad Tree
4 marks**Q3** ✓
Splay Tree
4 marks**Q4** ✓
Splay Trees
4 marks**Q5** ✓
Splay Tree
4 marks**Q6** ✓
Red-Black Tree
4 marks**Q7** ✓
Red-Black Tree**Q13 TRIE**

4 marks

In Trie, 'm' represents:

A. Number of nodes**B.** Length of the key**C.** Number of edges**D.** Height of tree Save Answer

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QUESTIONS

Q1
Tree Terminology
4 marks**Q2**
Quad Tree
4 marks**Q3**
Splay Tree
4 marks**Q4**
Splay Trees
4 marks**Q5**
Splay Tree
4 marks**Q6**
Red-Black Tree
4 marks**Q7****Q14 B-Tree**

4 marks

In a B-Tree of order 5, the maximum number of keys in a node is:

A. 4**B.** 5**C.** 6**D.** 3 Save Answer

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QUESTIONS

Q1
Tree Terminology
4 marks**Q2**
Quad Tree
4 marks**Q3**
Splay Tree
4 marks**Q4**
Splay Trees
4 marks**Q5**
Splay Tree
4 marks**Q6**
Red-Black Tree
4 marks**Q15 B-Tree**

4 marks

What is the minimum number of children for a non-root node in a B-Tree of order 'm'?

A. $m/2$ B. $m-1$ C. m D. $\lceil m/2 \rceil$  Save Answer

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QUESTIONS

Q1
Tree Terminology
4 marks**Q2**
Quad Tree
4 marks**Q3**
Splay Tree
4 marks**Q4**
Splay Trees
4 marks**Q5**
Splay Tree
4 marks**Q6**
Red-Black Tree**Q16 Search Tree**

4 marks

The main goal of using search trees is:

A. Increase complexity**B.** Reduce storage to zero**C.** Improve efficiency of data operations**D.** Avoid using memory Save Answer

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QUESTIONS

Q1 ✓
Tree Terminology
4 marks**Q2** ✓
Quad Tree
4 marks**Q3** ✓
Splay Tree
4 marks**Q4** ✓
Splay Trees
4 marks**Q5** ✓
Splay Tree
4 marks**Q6** ✓
Red-Black Tree
4 marks**Q7** ✓**Q17 AVL Trees**

4 marks

AVL trees are preferred over ordinary BST because they:

- A.** Use less memory always
- B.** Maintain balance for faster operations
- C.** Do not require insertion
- D.** Have no root node

 Save Answer

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QUESTIONS

Q1 ✓
Tree Terminology
4 marks**Q2** ✓
Quad Tree
4 marks**Q3** ✓
Splay Tree
4 marks**Q4** ✓
Splay Trees
4 marks**Q5** ✓
Splay Tree
4 marks**Q6** ✓
Red-Black Tree
4 marks**Q7** ✓
Red-Black Tree**Q18 Binary Search Trees**

4 marks

Which of the following is a common application of Binary Search Trees (BST)?

A. Database indexing**B.** Video editing**C.** Audio compression**D.** Network wiring Save Answer

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QUESTIONS

Q1
Tree Terminology
4 marks**Q2**
Quad Tree
4 marks**Q3**
Splay Tree
4 marks**Q4**
Splay Trees
4 marks**Q5**
Splay Tree
4 marks**Q6**
Red-Black Tree
4 marks**Q7**
Red-Black Tree**Q19 AVL Tree**

4 marks

Consider the AVL tree:

```
30
 /
10
 \
20
```

Which rotation is required?

A. Left Rotation**B.** Right Rotation**C.** Left Rotation + Right Rotation**D.** Right Rotation + Left Rotation Save Answer

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QUESTIONS

Q1
Tree Terminology
4 marks**Q2**
Quad Tree
4 marks**Q3**
Splay Tree
4 marks**Q4**
Splay Trees
4 marks**Q5**
Splay Tree
4 marks**Q6****Q20 AVL Tree**

4 marks

The maximum height difference allowed between left and right subtrees is:

B. 1**C. 2****D. 3** Save Answer

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QUESTIONS

Q1
Tree Terminology
4 marks**Q2**
Quad Tree
4 marks**Q3**
Splay Tree
4 marks**Q4**
Splay Trees
4 marks**Q5**
Splay Tree
4 marks**Q6**
Red-Black Tree
4 marks**Q21 Binary Search Tree**

4 marks

A data compression system uses a tree-based model to assign shorter codes to frequently used patterns. You need a self-adjusting BST where frequency is not stored explicitly. Which is most appropriate?

A. AVL Tree**B.** Threaded Binary Tree**C.** B-Tree**D.** Splay Tree

Save Answer

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QUESTIONS

Q1 ✓
 Tree Terminology
 4 marks

Q2 ✓
 Quad Tree
 4 marks

Q3 ✓
 Splay Tree
 4 marks

Q4 ✓
 Splay Trees
 4 marks

Q5 ✓
 Splay Tree
 4 marks

Q6 ✓
 Red-Black Tree
 4 marks

Q7 ✓
 Red-Black Tree

Q22 Tree Traversal

4 marks

Consider the binary tree:

A

/\

B C

/\

D E F

What is the Preorder traversal?

A. D B E A C F

B. A B D E C F

C. D E B F C A

D. B D E A C F

 Save Answer

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QUESTIONS

Q1
Tree Terminology
4 marks**Q2**
Quad Tree
4 marks**Q3**
Splay Tree
4 marks**Q4**
Splay Trees
4 marks**Q5**
Splay Tree
4 marks**Q6**
Red-Black Tree
4 marks**Q23 Tree Traversal**

4 marks

You are debugging a program that reconstructs the original root of a binary decision tree using traversal logs: Inorder and postorder. Given:
Inorder: D B E A F C Postorder: D E B F C A What is the root?

A. A**B.** B**C.** C**D.** D Save Answer

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QUESTIONS

Q1 ✓
Tree Terminology
4 marks**Q2** ✓
Quad Tree
4 marks**Q3** ✓
Splay Tree
4 marks**Q4** ✓
Splay Trees
4 marks**Q5** ✓
Splay Tree
4 marks**Q6** ✓
Red-Black Tree
4 marks**Q24 Binary Tree**

4 marks

The maximum number of nodes in a binary tree of height h is:A. h B. $2^h - 1$ C. $2^{(h+1)} - 1$ D. h^2  Save Answer

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QUESTIONS

Q1 ✓
Tree Terminology
4 marks**Q2** ✓
Quad Tree
4 marks**Q3** ✓
Splay Tree
4 marks**Q4** ✓
Splay Trees
4 marks**Q5** ✓
Splay Tree
4 marks**Q6** ✓
Red-Black Tree
4 marks**Q25 Tree Terminology**

4 marks

A path from a node to the root is formed by its:

☐ A. Descendants☒ B. Ancestors☐ C. Children☐ D. Leaves Save Answer